



Sediment characteristics affect thallus and rhizoid growth of the charophyte *Chara tomentosa* L.

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Abstract Shallow coastal bays and lagoons are important habitats supporting high biodiversity and many charophyte species. Charophyte habitats are particularly vulnerable to anthropogenic activity and restoring them has become a current management issue. To investigate the environmental requirements needed for charophyte site restorations, we tested how sediment characteristics affect the survival and growth of the key species *Chara tomentosa*. As *C. tomentosa* predominantly occurs in sheltered areas with organic muddy sediment, we hypothesised that sandier sediments would reduce its survival and growth. In aquaria, apical cuttings were planted in natural organic-rich sediment with three levels of sand addition (0, 40, and 80 vol%). *Chara tomentosa* survival was examined for 2.5 months, whereafter the growth of main thalli, developed side branches and rhizoids were analysed. *Chara tomentosa* survived and grew in all treatments, suggesting restoration through transplantation can be done by apical

cuttings. The probability of rhizoid development and the length of side branch and rhizoid growth were the highest in the natural sediment with no added sand, indicating limited growth potential in sediments with elevated sand content. Soft, organic-rich sediment was identified as the most suitable substrate for *C. tomentosa* transplants, but practical considerations may constrain its use in restoration efforts.

Keywords Macrophyte · Substrate · Nutrients · Grain size · Restoration · Baltic Sea

Introduction

Coastal bays and lagoons are highly productive ecosystems that sustain high biodiversity (Pérez-Ruzafa et al., 2024). These essential ecosystems include brackish charophyte habitats that support diverse invertebrate communities (Eveleens Maarse et al., 2021) and form feeding and nursery grounds for fish (Bučas et al., 2022) and water birds (Fox & Stipniece, 2024). Charophytes (Charales) are submerged macrophytes and they improve water quality in shallow bays through multiple mechanisms. Charophytes reduce water turbidity (Rosqvist et al., 2010) by competing with phytoplankton for nutrients (Pelechaty et al., 2006), binding sediment with their rhizoids, reducing resuspension and enabling particulate matter to settle by decreasing water movement (Madsen et al., 2001; Blindow et al., 2014).

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Charophytes consist of a photosynthetic thallus, and rhizoids attaching them to the soft sediment. Thallus branching (i.e. development of side branches) is very common (Blindow & Schütte, 2007; Schubert et al., 2024). Charophytes take up nutrients from both the sediment with their rhizoids and the water column with their thallus and can transport nutrients both ways between these sections (Vermeer et al., 2003; Wüstenberg et al., 2011). Charophytes can reproduce both sexually and asexually within an individual. Asexual reproduction can occur when attached or detached thallus segments contact the sediment and subsequently develop rhizoids from the thallus nodes (Skurzyński & Bociąg, 2011; Wüstenberg et al., 2011; Bociąg & Rekowska, 2012).

Environmental variables such as salinity, light and wave exposure strongly dictate the distribution of charophyte communities (Blindow & Schütte, 2007; Rosqvist et al., 2010; Bučas et al., 2019). Some charophyte species are tolerant to various salinities or occur solely in brackish water environments (Schubert & Blindow, 2004; Blindow & Schütte, 2007; Bučas et al., 2019). Insufficient light availability reduces growth in deeper waters, and the upper limit is often set by exposure to air or ice scouring (Blindow, 1992; Schubert & Blindow, 2004; Blindow & Schütte, 2007). Soft sediments, required by charophytes, generally vary in grain size, organic matter and nutrient content and these characteristics have a strong correlation with the degree of wave exposure (Madsen et al., 2001; Salmela et al., 2022; Keogh et al., 2025). In sheltered areas, larger fractions of fine-grained substrates like silt and clay and higher organic matter content of sediments are typical, because of higher local primary production, and hydrological conditions allowing fine-grained sediment deposition (Virtasalo et al., 2005). In sediments with higher organic matter content the total nutrient contents are typically higher (Puttonen et al., 2014; Winogrow & Pempkowiak, 2018).

In the northern Baltic Sea, there are many shallow bays and coastal lagoons. In these sheltered areas the largest charophyte species is *Chara tomentosa* Linnaeus. It can only grow in areas where mechanical stress caused by waves is low to moderate (Schubert & Blindow, 2004; Henricson et al., 2006; Forsblom et al., 2024). It has delicate rhizoids and in good environmental conditions thalli can reach lengths of 70–150 cm (Torn et al., 2006; Schubert et al.,

2024). The depth distribution range of *C. tomentosa* is 0.5–3.7 m (Schubert & Blindow, 2004; Schubert et al., 2024). It may form dense monospecific meadows or grow as mixed-species meadows with other charophyte species, including *Chara aspera* Willdenow, *Chara baltica* (Hartman) Bruzelius, *Chara canescens* Loiseleur, *Chara globularis* Thuiller and *Chara horrida* Wallman ex Wahlstedt, meaning they have partially overlapping niches (Herkül et al., 2018; Forsblom et al., 2024). Many of these charophyte species, for example *C. aspera*, can also be found on more exposed sandy coastal areas with moderate wave exposure (Schubert & Blindow, 2004; Bociąg & Rekowska, 2012; Forsblom et al., 2024).

The number and coverage of charophyte communities have been declining due to increasing anthropogenic impact (Blindow, 1992; Pitkänen et al., 2013). Charophytes are sensitive to turbulence and turbidity caused by human activity, including boating and dredging (Henricson et al., 2006; Blindow & Schütte, 2007; Wu et al., 2007; Hansen et al., 2019). Furthermore, charophytes have disappeared from many eutrophicated areas due to increasing water turbidity caused by particulate matter and phytoplankton, shading by periphyton and competition with other aquatic macrophyte species better adapted to eutrophic conditions (Blindow, 1992; Pitkänen et al., 2013; Nuñez et al., 2023).

To reduce ecosystem degradation and biodiversity loss recent global agreements for increased nature protection have been made, such as the Kunming-Montreal Global Biodiversity Framework (Secretariat of the Convention on Biological Diversity, 2022) and the EU's Biodiversity Strategy (European Commission, 2020). Additionally, the EU Nature Restoration Law (Regulation (EU) 2024/1991) aims to restore degraded habitats and species. Coastal lagoons are a priority habitat in the EU's Habitats Directive, Annex IV (Council Directive 92/43/EEC 1992) and have been classified as "Endangered" in the Baltic Sea (HELCOM, 2013). Further, sheltered charophyte dominated shallow soft bottoms have been classified as "Near Threatened" on the Baltic Sea scale (HELCOM, 2013) and "Vulnerable" on a national scale in Finland (Kotilainen et al., 2018).

Restoration projects aim to reverse ecosystem degradation and improve the status of habitats and species. For successful restoration, the stressor must be mitigated and the environmental requirements of

the species restored. In some cases, after removal of a stressor, species can spontaneously recolonise restoration areas (Balazy et al., 2024). Natural recolonisation can occur through germination from a local propagule bank or through dispersal from nearby populations. Alternatively, species can be actively transported from an existing population to an area where they once grew but have since disappeared. Charophytes can be transplanted by relocating whole individuals or only certain sections of them with or without the sediment from the source population site (Skurzyński & Bociąg, 2011; Wüstenberg et al., 2011; Faithfull et al., 2024). The aim of charophyte site restorations is the establishment of a new self-sustaining charophyte population. This requires the sediment at the restoration site to be suitable for the survival and growth of transplanted charophytes.

Most experiences of charophyte restorations are from freshwater (Dugdale et al., 2006; Knopik & Newman, 2018; Blindow et al., 2024). In the Baltic Sea, different restoration methods have been tested in a few small-scale trials, with varying results (Faithfull et al., 2024). The role of sediment characteristics for charophyte growth and survival is still poorly understood, especially in brackish water environments. In this aquarium experiment, the importance of sediment characteristics on *C. tomentosa* survival and growth was investigated by planting apical thallus cuttings in sediment with different levels of added sand.

The following research questions were addressed:

- (1) Do thallus fragments (apical cuttings) of *C. tomentosa* survive and develop rhizoids and side branches after planting?
- (2) How are thallus, side branch and rhizoid growth affected by the different levels of added sand in natural sieved sediment?

As *C. tomentosa* predominantly occurs in sheltered areas with organic muddy sediment, we hypothesised that sediment sand addition will negatively affect *C. tomentosa* growth and the development of rhizoids and side branches. Additionally, as charophyte collection depth has been shown to influence subsequent growth (Skurzyński & Bociąg, 2011), we assessed whether the survival and growth of *C. tomentosa* apical cuttings differed between two collection depths. These depths were selected to represent the upper

and lower limits of the depth range with the highest *C. tomentosa* cover at the collection site. We hypothesised that the relatively small difference in collection depth would not significantly affect the survival or growth of the planted apical cuttings.

Materials and Methods

To answer the research questions, a 2.5-month aquarium experiment was conducted at Husö Biological Station (Åbo Akademi University) in the Åland Islands, Finland in July–October 2023. In total 72 apical cuttings of *C. tomentosa*, collected from two depths, were planted across three sediment types in 12 aquaria. The collection depths were classified as shallow and deep. The sediment types were natural sieved sediment with 0, 40 or 80 volume percent (vol%) of added sand. Each sediment type was represented by 24 cuttings, with equal numbers originating from the shallow ($n = 12$) and deep ($n = 12$) collection depths. Each apical cutting was planted individually in a pot filled with the assigned sediment type. Each aquarium contained six pots, representing one replicate of each sediment type for both depth categories.

Water, sediment and charophyte apical cuttings used in the experiment originated from the nearby Svanviken bay (N 60.280620° E 19.837320°), which is a shallow, sheltered brackish water bay (salinity 4.7–4.9), with a large dense *C. tomentosa* meadow. The bay is surrounded by a traditional rural biotope combination of woodland and grassland with no buildings or jetties by the shoreline.

Aquarium system

The experiment was conducted in an aquarium system comprising 12 aquaria, each with outer dimensions of 35 × 30 × 30 cm (width × depth × height). Continuous water circulation was enabled by an overflow structure in the inside corner of each aquarium, resulting in a water volume of 24 l in each aquarium and flow rate of 0.83 ± 0.04 l/min (mean ± SE). All water used in the experiment was brackish water from the adjacent Svanviken bay filtered through a sand filter and a 40 µm mesh screen before entering the aquaria. Water coming from the aquaria through the overflow structure ran through a 200 µm mesh screen, a UV light filter (De Bary Aqua UV,

WF-25E) and a protein skimmer (Schuran Jetskim 120) before re-circulating back to the aquaria. Aquarium lights (EcoSmart Marine, Radion XR30w) were programmed with 16 h day length to reflect the local day length during the growing season. Light intensity at the bottom, middle and top of each aquarium was 380, 450, and 930 $\mu\text{mol}/\text{m}^2/\text{s}$, respectively (PAR sensor Apogee Instruments SQ-520). Water was added to the aquarium system to replace water loss from evaporation every 1–2 days and partially changed approximately every week to adjust salinity level and provide new nutrients to the circulating water. Water temperature, oxygen concentration, salinity and pH in all aquaria were measured once a week during daytime using a YSI ProSolo probe and a pH meter (Milwaukee MW806). The temperature of the water entering the aquaria was monitored with a logger (TinyTag) with a 60-min logging interval. In addition to in situ measurements, aquarium water was monitored by analysing total nitrogen (TN_w), total phosphorus (TP_w), dissolved nitrate and nitrite nitrogen ($\text{NO}_x\text{-N}$), dissolved phosphate phosphorus ($\text{PO}_4\text{-P}$), chlorophyll-*a* (Chl-*a*) and turbidity from water samples. At the beginning of the experiment, water nutrient samples were collected once from all aquaria, and at the middle and end of the experiment from four aquaria. Concentrations of TN_w , TP_w , $\text{NO}_x\text{-N}$ and $\text{PO}_4\text{-P}$ were analysed with a discrete analyzer (Seal Analytical AQ300). Samples for Chl-*a* analysis were collected from four aquaria on two occasions during the experiment, whilst turbidity samples were collected from four aquaria once. The Chl-*a* concentrations were measured using a ThermoSpectronic Aquamate spectrophotometer, and turbidity was measured using a Hach 2100P turbidimeter.

Sediment types used in experiment

Sediment was collected using an Ekman sampler from an area with a dense *C. tomentosa* meadow at a water depth of 0.5–0.8 m. Each sediment grab was run through a 1.0 mm sieve using brackish water to remove bigger sediment particles and organisms. The collected sediment and sediment water suspension was stored for a couple of days to allow the sediment to settle. Excess water was then removed and the sediment thoroughly mixed.

To adjust the sediment sand content, the sieved sediment was mixed with azooc sand (grain size

0.063–0.5 mm). Three sediment types AS00, AS40 and AS80, consisting of natural, sieved sediment mixed with sand in volume percentages of 0, 40, and 80, respectively, were used in the experiment. Samples of each sediment type (AS00, AS40 and AS80) were taken for sediment analyses. The water content and organic matter content as loss-on-ignition (LOI) of each sediment type ($n = 3$ per sediment type) were analysed by drying the samples at 105 °C for 24 h and then combusting at 550 °C for 4 h. Sediment grain size analysis ($n = 1$ per sediment type) was done at the Geology Department at Åbo Akademi University, using the hydrometer method (Ashworth et al., 2001). Organic matter was removed from the AS00 sample before the hydrometer analysis by oxidation with 30% hydrogen peroxide. Organic matter may cause flocculation resulting in underestimation of fine particles, but when the sediment organic carbon content is less than 2%, the impact to the estimates of fine particles is negligible (Jensen et al., 2017). The sediments with added sand were not pretreated for the analysis due to their low organic matter and thus low organic carbon content.

Total nitrogen, total phosphorus and total carbon contents of the sediments (TN_s , TP_s and TC , respectively) were analysed from freeze-dried samples ($n = 1$ per sediment type) at Tvärminne Zoological Station, University of Helsinki. Total phosphorus was analysed following the procedure for particulate phosphorus determination in Solorzano & Sharp (1980), except using 1 M HCL instead of 0.2 M HCL in the hydrolysis. The mixed reagent for TP_s analysis was prepared by mixing sulphuric acid, ammonium molybdate, ascorbic acid and potassium tartrate solutions (Murphy & Riley, 1962). Phosphorus in the extant was analysed as phosphate by detecting the optical density of the molybdenum blue complexes from the sample solution (Spilling, 1980) with an automatic photometric analyser (Thermo Scientific Aquakem 250 analyser). Total nitrogen and total carbon content were analysed with an automatic C/N analyser (Europa Scientific ANCA 20/20 stable isotope analyser with a Sercon GSL sample preparation system) according to the methods described in Spilling (1980).

To prepare the aquaria for planting *C. tomentosa* apical cuttings, 24 plastic pots (diameter 7 cm, volume 0.15 l) were filled with each sediment type, resulting in a total of 72 pots. Six pots, including two

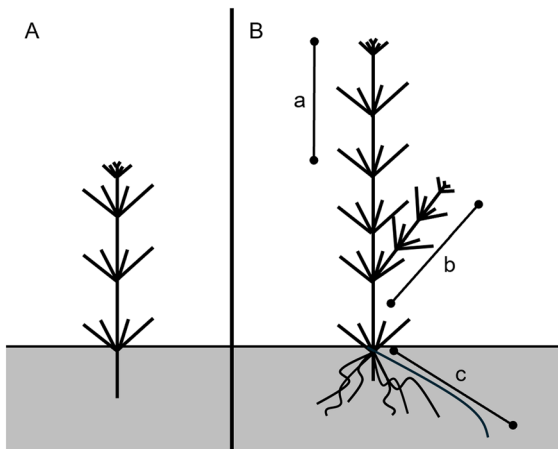


Fig. 1 Illustration of an apical cutting at the **A** beginning and **B** end of the experiment. Apical cutting growth was measured as **a** main thallus growth, which was the difference between the main thallus length at the beginning and end of the experiment, **b** total length of all side branches and **c** the maximum length of rhizoids. The shaded area illustrates sediment and the unshaded area water above the sediment

of each sediment type, were placed in each aquarium. The 12 aquaria were filled with brackish water prior to the careful placement of the pots to minimize sediment resuspension.

Collection of charophytes, measuring and planting of apical cuttings

Chara tomentosa were collected from the same bay as the sediment. In July 2023, 24 whole *C. tomentosa* charophytes, each with at least three apical tips and growing at least 0.5 m apart, were hand-collected by snorkelling at two depths, shallow (0.58–0.68 m) and deep (0.95–1.12 m). In the laboratory, the charophytes were photographed in a shallow wide container filled with brackish water next to a ruler. Three apical cuttings were cut from each charophyte. The apical cuttings consisted of an apical tip, three internodes and a section of a fourth internode to be pushed into the sediment (Fig. 1). Apical cuttings did not have any side branches or rhizoids. The length of each apical cutting was measured to the nearest millimetre. In total, 72 apical cuttings were used in the experiment.

The three apical cuttings obtained from a single charophyte were planted in the same aquarium, one in each sediment type. In total, six apical cuttings were

planted per aquarium, with three originating from the shallow area and three from the deep area. One apical cutting was planted in each pot by pushing it into the sediment to the depth where the lowest branchlets and node were touching the surface of the sediment. Due to the sediment type AS80 being coarser and denser, a wooden stick was pushed vertically into AS80 sediment pots and removed right before planting to prevent damage to the apical cuttings. Apical cuttings were planted within 12 h of charophyte collection and kept in brackish water whenever possible during the processing.

The experiment lasted 2.5 months. Any apical cutting that detached from the sediment before the end of the experiment was removed from the aquarium. At the end of the experiment, the pots were removed from the aquaria, and the apical cuttings and sediment were removed from the pots. The apical cuttings were then carefully rinsed and measured.

Three growth variables were determined for the apical cuttings: main thallus growth, side branch growth and rhizoid growth. Main thallus growth was calculated as the difference between the beginning and end length of the main thallus, measured to the nearest millimetre from the lowest node to the tip of the main thallus (the section above the sediment, Fig. 1). Due to unexpected thallus decomposition within the sediment, photographs taken at the beginning of the experiment were analysed using ImageJ software (Version 1.54 g, Java 1.8.0_345; Schneider et al., 2012) to determine the initial lengths of the main thalli. Out of the 72 planted apical cuttings, this measurement could reliably be determined for 32 apical cuttings included in the main thallus analyses. Photographs were taken of each whole charophyte instead of each apical cutting, and identification of many of the apical cuttings from the photographs alone was not possible. Out of these 32 apical cuttings, 15 were collected from the shallow depth and 17 were collected from the deeper depth. From the shallow depth five of these apical cuttings were grown in the sediment type AS00, five in AS40 and five in AS80. From the deeper depth five of these apical cuttings were grown in the sediment type AS00, six in AS40 and six in AS80.

All apical cuttings were visually examined to determine if they had developed side branches or rhizoids during the experiment. The length of each side branch and the length of the longest rhizoid per

apical cutting was measured to the nearest millimetre (Fig. 1). Hereafter the side branch growth is the total length (sum) of all side branches per apical cutting, and the rhizoid growth is the length of the longest rhizoid per apical cutting. The maximum length of rhizoids was deemed as the most reliable measurement for the rhizoid growth as measuring each rhizoid or removing all attached sediment particles would have damaged the rhizoids and provided unreliable total length and biomass data.

Statistics

Generalized mixed effect models (GLMM) were applied to assess the effect of sediment characteristics on *C. tomentosa* growth (main thalli, side branches and rhizoids). The volume percentage of added sand was included as a predictor and each model included aquarium as a random intercept. The effect of collection depth was tested by including it as a covariate in the model but dropped due to non-significant effects on growth. Due to zero inflation in the growth data, a hurdle model approach was applied. Hurdle models consisted of two steps: a binomial model and a truncated positive gamma model. For the binomial model, the growth data were converted into a binomial format by keeping all 0 values as 0 and replacing all values > 0 with the value 1. The binomial model was run using a binomial distribution and logit-link function. The output from the binomial models was the predicted probability for apical cuttings to exhibit main thallus growth or develop side branches or rhizoids (presence/absence at the end of the experiment) in each sediment type. In the truncated positive gamma model, the original unconverted growth data were used, datapoints with 0 growth were excluded, and the model was run using only the positive values (growth > 0 mm). The model was run using a gamma distribution and log-link function, and the output was the predicted length of the main thallus, side branch and rhizoid growth in millimetres in each sediment type. The prediction values from the hurdle models were acquired by combining the binomial and gamma models by multiplying the modelled values. Hurdle models were built in R Studio (R Core Team 2024, version 4.4.0) using the `glmmTMB` function in the `glmmTMB` package (Brooks et al., 2017). The random effect of aquarium was assessed using the `ranef` function in the `glmmTMB` package. The fit of the

models was assessed using the Akaike information criterion.

All measured sediment characteristics (sediment water content, LOI, nutrient contents and average grain size) were highly correlated with the volume percentage of added sand (Pearson correlation: $r > 0.9$) and thus not included in the hurdle models. All apical cuttings in one of the aquaria grew very poorly. These were confirmed as outliers in the dataset by the random effect analyses and subsequently removed from downstream analyses, except for when calculating the total survival of apical cuttings. Two out of all of the planted apical cuttings detached during the experiment. These two were only considered when calculating the survival of all apical cuttings and excluded from all other analyses. This means that in total 32 apical cuttings were included in the main thallus analyses and 64 apical cuttings were included in the side branch and rhizoid analyses. Negative growth values in four of 32 analysed main thalli (growth < 0 mm) were considered to be caused by the error margin of the measurement techniques and converted to zero (growth = 0 mm).

The relationship between side branch growth and rhizoid growth per apical cutting was assessed using Spearman's correlation to determine if side branches and rhizoids developed on the same apical cuttings. In this analysis, the data were pooled across sediment types.

Results

Sediment and aquarium water

The organic matter and nutrient contents were higher and the average grain size was smaller in the sediment type AS00 with no added sand compared to the sediment types with added sand (Table 1). The water content was higher in AS00 compared to AS40 and AS80. During the experiment, water parameters were stable and variations between the aquaria were minor (Online Resource 1).

Survival and growth of apical cuttings

Out of all planted apical cuttings ($n = 72$), 97% survived until the end of the experiment. The two detached apical cuttings were planted in sediment

Table 1 Properties of the sediment types. Abbreviations AS00, AS40, and AS80 indicate sediment types of natural sieved sediment with 0, 40, and 80 volume percentage (vol%) added sand

Sedi- ment type	Water content (%) ww; mean $\pm$ SE)	LOI (% dw; mean $\pm$ SE)	Clay (% dw)	Silt (% dw)	Sand (% dw)	TP _s (μ g/g)	TN _s (μ g/g)	TC (μ g/g)
AS00	81.7 $\pm$ 0.2	12.08 $\pm$ 0.11	39	55	6	808.3	6835.8	59,204.4
AS40	51.2 $\pm$ 1.1	2.58 $\pm$ 0.12	7	11	82	298.2	1068.6	10,793.1
AS80	25.6 $\pm$ 0.6	0.66 $\pm$ 0.00	5	2	93	277.5	< 300 (120)	1821.7

Water content is expressed as a percentage of wet weight. Organic matter content is estimated as loss-on-ignition (LOI) and expressed as a percentage of dry weight. Total nitrogen, total phosphorus and total carbon content of the sediment types, abbreviated as TP_s, TN_s and TC, respectively, are expressed as μ g per g of dry sediment. Grain size distribution is expressed as percentages of dry weight per the diameter of mineral sediment particles of clay (< 2 μ m), silt (2–63 μ m) and sand (> 63 μ m). Detection limit for TN_s was 300 μ g/g, measured values under detection limit indicated in brackets. Standard error is indicated by SE. Water content and LOI were analysed from three samples per sediment type, grain size distribution and nutrient contents from one sample per sediment type. Note that the sand addition was measured in volume and the results from sediment analyses are expressed as weight

types AS00 and AS80 in two different aquaria. Planted apical cuttings exhibited main thallus growth and developed side branches and rhizoids in all three sediment types, but with varying results.

The proportion of apical cuttings that exhibited main thallus growth was similar in all treatments (Table 2) and there was no statistical difference in main thallus growth between the three sediment types (Table 3). The observed length of main thallus growth was 41% lower in AS40 and 35% lower in AS80 compared to AS00, when only positive growth values (growth > 0 mm) in each sediment type were included in the calculations (Table 2; Fig. 2). Collection depth did not influence the probability of main thalli to exhibit growth ($P = 0.515$) or the length of main thallus growth ($P = 0.162$).

The addition of sand did not affect the probability of side branch development (Table 3, binomial model). Observed length of side branch growth was 58% lower in AS40 and 90% lower in AS80 compared to AS00, when only positive growth values (growth > 0 mm) in each sediment type were included in the calculations (Table 2; Fig. 2). Based on the hurdle model, increasing volume percentage of added sand had a significant negative effect on the length of side branch growth ($P < 0.001$; Table 3, gamma model; Fig. 2). The collection depth of *C. tomentosa* had no effect on the probability ($P = 0.955$) or the length of side branch growth ($P = 0.818$).

Based on the observed and modelled values, the probability of rhizoid development was higher, and the rhizoids grew longer in natural sieved sediment compared to the sediment types with added sand

(Table 2 and 3). The proportion of individuals that developed rhizoids in AS00 was 81% (Table 2), and it was 32% lower in AS40 and 53% lower in AS80 relative to AS00. The corresponding values for the length of rhizoid growth were 28 and 73% lower, respectively, when only positive growth values (growth > 0 mm) were included in the calculations. The collection depth had no effect on the probability ($P = 0.576$) or the length of rhizoid growth ($P = 0.214$).

There was a positive correlation between side branch growth and rhizoid growth per apical cutting (Spearman correlation: $\rho = 0.775$, $P < 0.001$; Fig. 3). In 28% of the apical cuttings the length of both the side branch and rhizoid growth was 0 mm, 25% of the apical cuttings only developed either side branches or rhizoids and 47% of the apical cuttings developed both. The mean of the total length of side branches growing on apical cuttings with rhizoids (rhizoid length > 0 mm) was 42.32 mm, and the corresponding mean on apical cuttings with no rhizoids (rhizoid length = 0 mm) was 2.15 mm.

Discussion

Sediment sand additions had negative effects on *C. tomentosa* growth in our brackish aquarium experiment. Side branches and rhizoids developed and grew in all sediment types, but with varying results. Elevated sediment sand content did not influence the probability of side branch development, but the probability for rhizoid development decreased with increasing sediment sand content. Likewise, the

Table 2 The observed growth of main thalli, side branches and rhizoids on apical cuttings during the experiment

Sediment type	Main thallus growth			Side branch growth			Rhizoid growth					
	Proportion of apical cuttings (n)	Proportion of apical cuttings (%)	Length (n)	Length (mm; mean $\pm$ SE)	Proportion of apical cuttings (n)	Proportion of apical cuttings (%)	Length (n)	Length (mm; mean $\pm$ SE)	Proportion of apical cuttings (n)	Proportion of apical cuttings (%)	Length (n)	Length (mm; mean $\pm$ SE)
AS00	10	80	8	15.88 $\pm$ 1.84	21	71	15	74.33 $\pm$ 18.58	21	81	17	57.76 $\pm$ 8.06
AS40	11	82	9	9.33 $\pm$ 1.46	22	64	14	31.21 $\pm$ 8.34	22	55	12	41.42 $\pm$ 8.57
AS80	11	82	9	10.33 $\pm$ 2.27	21	48	10	7.20 $\pm$ 1.25	21	38	8	15.88 $\pm$ 4.55

Percentages indicate the proportion of apical cuttings that exhibited main thallus, side branch or rhizoid growth. Length (mm; mean $\pm$ SE) indicates mean length in millimetres calculated for positive growth values (growth > 0 mm). Abbreviations AS00, AS40 and AS80 indicate sediment types with 0, 40 and 80 volume percentage sand addition. Main thallus growth is the difference between main thallus length at the beginning and end of the experiment. Side branch growth is the total length of all side branches per apical cutting. Rhizoid growth is the length of the longest rhizoid per apical cutting. Abbreviation SE indicates standard error. Number of observations (n) refers to the number of apical cuttings included in the calculations

length of the side branch and rhizoid growth was longer in the sediment with no added sand compared to sediments with higher sand contents. The correlation between side branch and rhizoid growth in individual apical cuttings was high. Almost all apical cuttings survived throughout the whole experiment.

Sand addition to natural sieved sediment changed multiple variables, including sediment grain size, organic matter, total nutrient and water contents, and changes in these variables are often linked in natural ecosystems (Puttonen et al., 2014; Asmala et al., 2017; Winogradow & Pempkowiak, 2018). We observed negative associations with sediment nutrient (TP_s, TN_s, TC) and water content with increasing sand content (Table 1). The quantity of nutrients in sediments has ecological consequences, with nutrient limitation often reducing growth rate in macrophytes (Wüstenberg et al., 2011). We found that rhizoid development was more probable and side branches and rhizoids grew longer in sediment with no added sand, which is consistent with previous findings of high sediment sand content inhibiting charophyte growth (Matheson et al., 2005). The mean concentrations of dissolved NO_x-N (1.9 µg/l) and PO₄-P (3.9 µg/l) in the aquarium water were low (Online Resource 1) and represent similar values commonly measured in the Baltic Sea brackish water bays during seasonal minimum values (Holmboe et al., 1999; Nyman, 2021). Consequently, although it was not explicitly tested for, we suggest that the higher nutrient pool in the lower sand content treatments was likely one factor promoting the increased growth. This assumption is supported by a laboratory experiment that showed increased growth of *Chara hispida* Linnaeus in phosphate-enriched sand compared to sand without nutrient addition (Wüstenberg et al., 2011). The weaker rhizoid growth with increasing sediment sand content may have also been linked to the physical differences in grain size and water content. The rhizoids of *C. tomentosa* are delicate meaning that it could be more challenging and energy-demanding for them to grow amongst heavier and denser sandy sediments due to mechanical impedance (Correa et al., 2019), than in fine-grained, organic-rich sediments with high water content (Puttonen et al., 2026).

The fragile nature of rhizoids may have also contributed to some breakage when sediment was gently rinsed from the apical cuttings at the end of the experiment. This could have introduced error into the

Table 3 Effect of sediment sand addition on the probability and length of main thallus, side branch and rhizoid growth as the output of hurdle models

Model	Response variable	Number of observations	Estimate	SE	z value	P value
Binomial	Main thallus growth (%)	32	0.00131	0.0146	0.0900	0.929
Gamma	Main thallus growth (mm)	26	− 0.00523	0.00322	− 1.623	0.105
Binomial	Side branch growth (%)	64	− 0.0128	0.00818	− 1.563	0.118
Gamma	Side branch growth (mm)	39	− 0.0278	0.00510	− 5.443	5.23E−08
Binomial	Rhizoid growth (%)	64	− 0.0285	0.0104	− 2.757	0.00583
Gamma	Rhizoid growth (mm)	37	− 0.0158	0.00437	− 3.605	3.12E−04

Main thallus growth is the difference between main thallus length at the beginning and end of the experiment. Side branch growth is the total length of all side branches per apical cutting. Rhizoid growth is the length of the longest rhizoid per apical cutting. The results of binomial models are on a logit scale and gamma models on a log scale. Model estimate with standard error (SE) and test-statistic (z). Significant *P* values ($P < 0.05$) are indicated in bold

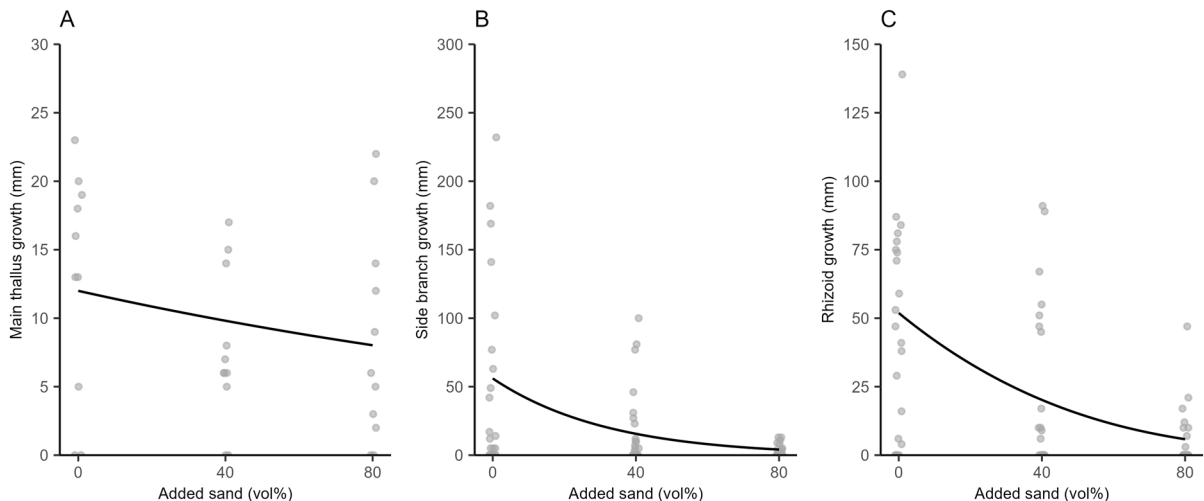


Fig. 2 Observed and modelled **A** main thallus, **B** side branch and **C** rhizoid growth in millimetres. Observed values are illustrated as grey points. Black curves represent the combined model prediction from a two-part hurdle model consisting of a GLMM with a binomial error distribution and a GLMM with a gamma distribution. Main thallus growth is the difference between main thallus length at the beginning and end of the experiment. Side branch growth is the total length of all side branches per apical cutting and rhizoid growth is the

length of the longest rhizoid per apical cutting measured at the end of the experiment. X-axis illustrates the volume percentage (vol%) of added sand in the sediment. Models with *P* values < 0.05 are considered significant. (A) binomial model $P = 0.929$; gamma model $P = 0.105$ (B) binomial model $P = 0.118$; gamma model $P = 5.23\text{E}−08$ (C) binomial model $P = 0.00583$; gamma model $P = 3.12\text{E}−04$. Note the differing scales on y-axis

measurements of the length of the longest rhizoid per apical cutting. This potential limitation may be worth considering when interpreting the results. Nevertheless, the findings were clear and were further supported by visual observations of rhizoid development as both the number of rhizoids and their total length appeared to be positively related to the length of the longest rhizoid.

The relationship of more probable and longer rhizoid growth in finer sediment with higher nutrient contents may also have further promoted the growth of side branches in this sediment type. Well-developed rhizoids increase the nutrient uptake potential and likelihood of a charophyte staying attached (Vermeer et al., 2003) and thus support additional growth of the thallus. This can be seen through the positive

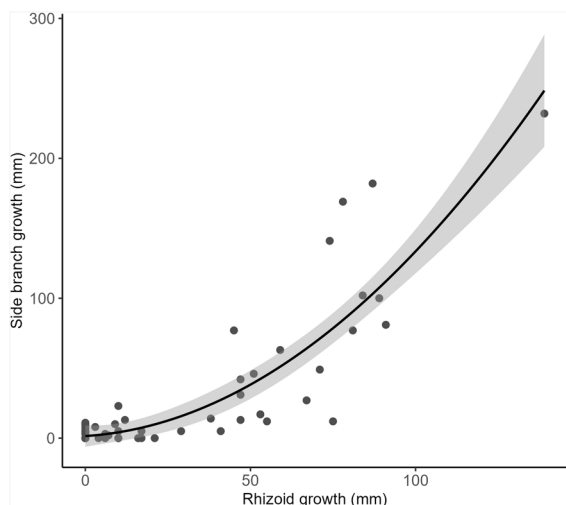


Fig. 3 The Spearman correlation between side branch growth and rhizoid growth per apical cutting in millimetres ($\rho = 0.775$, $P < 0.001$) with a fitted quadratic model regression curve ($R^2 = 0.779$). The shaded area indicates standard error. Side branch growth is the total length of all side branches per apical cutting and rhizoid growth is the length of the longest rhizoid per apical cutting measured at the end of the experiment

correlation between rhizoid and side branch growth. The majority of apical cuttings allocated energy into developing both side branches and rhizoids, meaning they allocated resources to increase both photosynthetic activity and nutrient uptake. However, in our experimental setup, it could not be determined whether the initiation of rhizoid and side branch growth occurred in asynchronicity or synchronicity. Regardless, we suggest that the development of extensive rhizoids supported the development and growth of side branches.

In charophytes, new growth of the thallus can reduce the negative effects of shading by a physical layer, such as epiphytic algae or sedimentation, in immediate contact with the thallus. Shading can reduce photosynthetic activity, leading to lower growth rates and survival (Paar et al., 2021). New growth can occur from nodal tissues on both apical and non-apical parts of the thallus, even from one-node thallus fragments without an apical tip (Skurzyński & Bociąg, 2011). In our experiment, thallus growth occurred at both the apical tip of the main thallus and as the development of side branches from the non-apical thallus nodes. Given

also the rhizoid growth during the experiment, our study shows that apical cuttings of *C. tomentosa* can develop the necessary biological structures needed to promote their individual survival. This poses a less-invasive method for active restoration where only a fragment of the thallus is transplanted rather than the whole charophyte.

There has been a decline of charophyte habitats due to anthropogenic stressors, and knowledge on environmental requirements is essential for successful restoration (Blindow, 1992). In this study, we showed that elevated sediment sand content affects charophyte growth, and our results suggest that sediment characteristics are important to consider when planning restoration efforts. Regarding transplanting, different sediment types with low and high sand content have both practical and impractical characteristics. Physical disturbance and altered water movement easily resuspend very fine-grained sediments with high organic matter and water content. This can make practical restoration actions in environments with fine-grained sediment difficult due to temporary loss of visibility in the water column (Faithfull et al., 2024). Additionally, resuspended particles may cause shading, and when deposited in large quantities, have detrimental effects on charophytes (Henricson et al., 2006). Coarser and denser sediments with higher sand content do not resuspend as easily, but inserting a transplant into the substrate can be challenging without damaging the charophyte. Further, our results suggest that the development and growth of rhizoids may be hampered in coarser sediments, and subsequently, the overall long-term survival of transplants is uncertain, consequently affecting restoration success.

In suboptimal conditions, transplanted charophytes may not establish properly and are potentially more easily outcompeted by other species better adapted to the conditions in the area (Blindow, 1992; Rosqvist et al., 2010). Based on restoration work in freshwater and brackish water sites, transplanting charophytes is expensive and the success of these efforts is relatively low (Blindow et al., 2024; Faithfull et al., 2024). Hence, transplanting charophytes in the most suitable conditions should be prioritised to increase the chance of success. The trade-offs of using different sediment types should be thoroughly evaluated, including site-specificity

and the sediment types naturally occurring and readily available in the restoration areas.

Our results indicate that *C. tomentosa* collected from a brackish water site can be cultivated from thallus fragments without initial rhizoids, which has also been shown to be possible for *C. hispida* collected from an artificial pond (Wüstenberg et al., 2011). Further, our results show that the collection depth does not affect the survival or growth of *C. tomentosa*, suggesting they can be collected from varying depths (approximately 0.5–1.0 m) for transplanting. Both of these findings can make practical restoration work easier, lower the harvesting pressure on source populations and help alleviate the risk of failed restoration attempts contributing to decline of this species. Future research to support restoration of *C. tomentosa* should include studies on other additional environmental variables such as turbidity and light conditions, effective reproduction strategies such as exploring methods of breaking oospore dormancy to facilitate germination and assessing growth and establishment of transplanted thallus fragments in field conditions. Genetic studies on source populations would clarify connectivity patterns and aid choosing source populations for restoration trials.

Conclusions

This study implies that *C. tomentosa* apical cuttings remain viable after replanting, but increasing sediment sand content reduces the likelihood of rhizoid development, presumably affecting the rigidity of attachment to sediment and nutrient uptake capacity. Although main thallus growth did not differ between sediment types, side branch and rhizoid growth were both hampered in sand-enriched substrates relative to natural sediment, which contained higher levels of organic matter, nutrients and finer particles. Regardless, some apical cuttings grew in the sediment types with elevated sand content, thus sediments with moderate sand content could be considered as a substrate option in restoration projects. This is valuable information for restoration implementation, since the natural *C. tomentosa* sediment is generally very soft, which can make the practical transplanting work difficult. The rhizoid development and thallus growth of the apical cuttings confirm that restoration through transplantation does not need to use whole

individuals, as clonal propagation poses a feasible restoration method. This, combined with the finding that collection depth had no effect on the growth of apical cuttings, meaning *C. tomentosa* could be harvested from a range of depths (approximately 0.5–1.0 m) for transplanting, can reduce the harvesting pressure on source populations.

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Data availability Data are publicly available in a data repository (<https://doi.org/10.5281/zenodo.19599015>).

Declarations

Competing interest The authors have no relevant financial or non-financial interests to disclose.

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